
Dataset 4 — LDA Domain Comparison + Logistic Regression

Heart Disease Risk Classification (n = 320 patients)

Research Question: Do lifestyle behavioral factors (Exercise, Stress, Smoking) or clinical biomarkers (Age, BMI, Blood Pressure, Cholesterol) provide greater discriminating power between cardiac risk groups? Which domain — or their combination — yields optimal classification?

Methods Used:

- Linear Discriminant Analysis — Lifestyle Domain Only (Model 1)
- Linear Discriminant Analysis — Clinical Biomarker Domain Only (Model 2)
- Linear Discriminant Analysis — Full Combined Model (Model 3)
- Logistic Regression — Full Model Comparison
- 10-Fold Cross-Validated Accuracy for each model

Dataset:

- N: 320 patients
- Target: Heart_Disease_Group (High Risk / Low Risk)
- Lifestyle predictors: Exercise_Hours_Per_Week, Stress_Level, Smoking_Years
- Clinical predictors: Age, BMI, Blood_Pressure, Cholesterol
- Train/Test split: 75% / 25%

Submitted by: Lana Gidan

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Part A — Variable-Level Discrimination (t-tests)

Before building discriminant models, individual t-tests identify which variables show significant mean differences between High Risk and Low Risk groups. Cohen's d quantifies effect size (0.20=small, 0.50=medium, 0.80=large).

Part B — LDA Domain Comparison

Three separate LDA models are built to directly compare the discriminating power of lifestyle versus clinical predictors. This reveals which domain of risk factors the model relies on most.

Metric	Lifestyle	Clinical	Full Model	Best
Accuracy	62.50%	67.50%	76.25%	Full ✓
AUC-ROC	0.6681	0.7244	0.8350	Full ✓
Sensitivity	62.50%	77.50%	82.50%	Full ✓
Specificity	62.50%	57.50%	70.00%	Full ✓
F1 Score	0.6250	0.7045	0.7765	Full ✓
CV Accuracy	64.69%	66.88%	75.00%	Full ✓

Domain Comparison — Significance Commentary

Winner: Full domain. Lifestyle Model: Accuracy = 62.50%, AUC = 0.6681. Clinical Model: Accuracy = 67.50%, AUC = 0.7244. Full Model: Accuracy = 76.25%, AUC = 0.8350. The Full model wins on the most metrics, suggesting that the full combination of lifestyle and clinical factors is most informative — no single domain alone captures the complete risk picture.

Part C — Full LDA Model Details

C1. Discriminant Function Coefficients

Standardized discriminant function coefficients indicate each variable's unique contribution to the discriminant function, controlling for other predictors. Larger |coefficient| = more powerful discriminator.

Rank	Variable	Std Coef	Domain	Strength
1	Exercise_Hours_Per_Week	+0.9567	Lifestyle	Strong
2	Stress_Level	-0.9320	Lifestyle	Strong
3	Blood_Pressure	-0.8570	Clinical	Strong
4	Cholesterol	-0.8525	Clinical	Strong
5	BMI	-0.7689	Clinical	Strong

6	Age	-0.6091	Clinical	Strong
7	Smoking_Years	-0.3438	Lifestyle	Moderate

C2. Structure Matrix (Canonical Loadings)

Structure coefficients are correlations between each original variable and the discriminant function. Unlike coefficients, they are not affected by multicollinearity and are considered more reliable for interpretation.

Rank	Variable	r (structure)	Domain	Strength
1	Exercise_Hours_Per_Week	+0.4726	Lifestyle	***
2	Stress_Level	-0.4485	Lifestyle	***
3	Blood_Pressure	-0.3973	Clinical	***
4	BMI	-0.3933	Clinical	***
5	Age	-0.3526	Clinical	***
6	Cholesterol	-0.2851	Clinical	**
7	Smoking_Years	-0.1700	Lifestyle	*

C3. LDA vs Logistic Regression

Metric	Full LDA	Logistic Regression	Winner
Accuracy	76.25%	76.25%	Tie
AUC-ROC	0.8350	0.8337	LDA ✓
Sensitivity	82.50%	82.50%	Tie
CV Accuracy	75.00%	75.00%	Tie

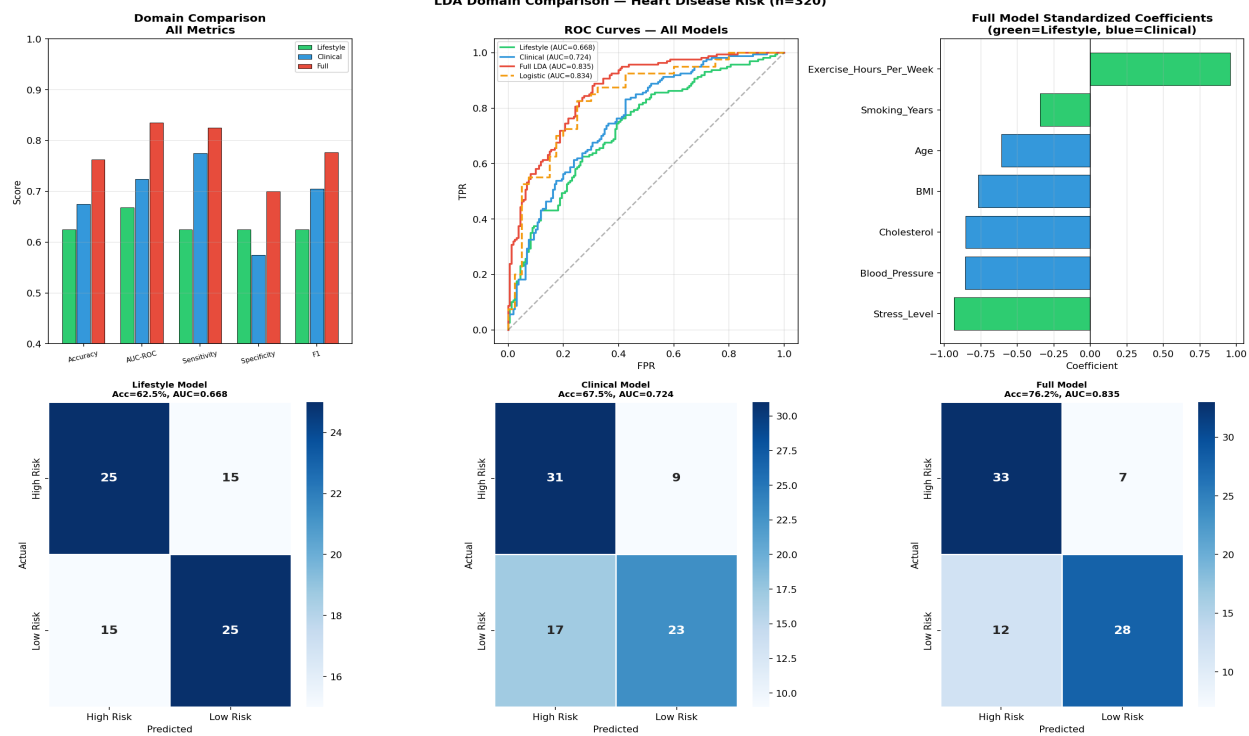


Figure 1. Left: Domain comparison bar chart. Center: ROC curves all models. Right: Full model standardized coefficients. Bottom: Confusion matrices.